

Duality Theory

• Symmetry

$$\begin{array}{llll} \max C^T x & \textcircled{1} & -\min -C^T x & \textcircled{+} & -\min C^T y \\ \text{s.t. } Ax \leq b & \Leftrightarrow & -Ax \geq -b & \Leftrightarrow & \text{s.t. } (A^T)^T y \geq -b \\ x \geq 0 & & x \geq 0 & & y \leq 0 \end{array}$$

$$\begin{array}{llll} \textcircled{1} \downarrow \text{dual} & & & \\ \min b^T \pi & \textcircled{2} & -\max -b^T \pi & \textcircled{3} \nearrow \text{dual} \\ \text{s.t. } A^T \pi \geq c & \Leftrightarrow & \text{s.t. } A^T \pi \geq c \\ \pi \geq 0 & & \pi \geq 0 & \end{array}$$

Thm The dual of the dual is the primal

Therefore, we can also read the table in last lecture from right to left

e.g.: The dual problem of

$$\begin{array}{ll} \min Z = 3x_1 + 2x_2 - 4x_3 \\ \text{s.t. } 5x_1 - 7x_2 + x_3 \geq 12 \\ x_1 - x_2 + 2x_3 = 18 \\ 2x_1 - x_3 \leq 6 \\ x_1, x_2, x_3 \geq 0 \end{array}$$

is

$$\begin{array}{ll} \max Z = 12\pi_1 + 18\pi_2 + 6\pi_3 \\ \text{s.t. } 5\pi_1 + \pi_2 + 2\pi_3 \leq 3 \\ -7\pi_1 - \pi_2 \leq 2 \\ \pi_1 + 2\pi_2 - \pi_3 \leq -4 \\ \pi_1 \geq 0, \pi_3 \leq 0 \end{array}$$

maximization	maximization
Constraints	Variables
$\leq$	$\geq$
$\geq$	$\leq$
$=$	unrestricted
Variables	Constraints
$\geq$	$\geq$
$\leq$	$\leq$
unrestricted	$=$

• Weak duality

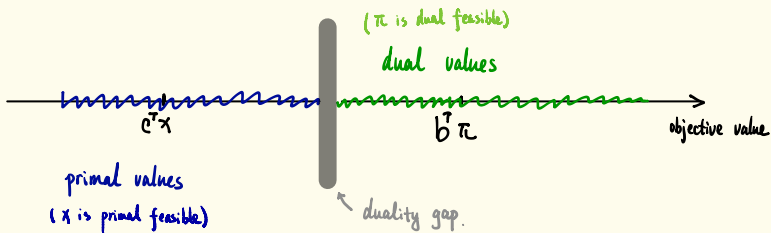
Consider the following primal-dual pair

$$\begin{array}{ll} \max Z = C^T \bar{x} \\ \text{s.t. } A\bar{x} \leq b \quad (P) \\ \bar{x} \geq 0 \end{array} \quad \longleftrightarrow \quad \begin{array}{ll} \min Z = b^T \bar{\pi} \\ \text{s.t. } A^T \bar{\pi} \geq C \quad (D) \\ \bar{\pi} \geq 0 \end{array}$$

Thm If $\bar{x}$ is a feasible solution to the maximization problem (P) and $\bar{\pi}$ is a feasible solution to the minimization problem (D), then $C^T \bar{x} \leq b^T \bar{\pi}$

$$\begin{array}{c} \text{pf} \quad C^T \bar{x} \leq (A^T \bar{\pi})^T \bar{x} = \bar{\pi}^T A \bar{x} \leq \bar{\pi}^T b \\ \quad \quad \quad \uparrow \quad \quad \quad \uparrow \quad \quad \quad \uparrow \\ \quad \quad \quad A^T \bar{\pi} \geq C \quad \text{property of} \quad A \bar{x} \leq b \\ \quad \quad \quad \text{and } \bar{x} \geq 0 \quad \text{transpose} \quad \text{and } \bar{\pi} \geq 0 \end{array}$$

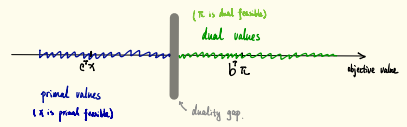
Cor If the primal problem is unbounded, then the dual problem is infeasible



Remark: The converse of the corollary is not true.

$$\begin{array}{ll} \text{eg} \quad \max x_1 + 3x_2 \\ \text{s.t. } x_1 + 2x_2 = 1 \quad (P) \\ \quad \quad x_1 + 2x_2 = 2 \end{array} \quad \xleftrightarrow{\text{dual}} \quad \begin{array}{ll} \min \pi_1 + 2\pi_2 \\ \text{s.t. } \pi_1 + \pi_2 = 1 \quad (D) \\ \quad \quad 2\pi_1 + 2\pi_2 = 3 \end{array}$$

Both (P) and (D) are infeasible



Cor: If $\bar{x}$ is feasible to (P), and $\bar{\pi}$ is feasible to (D), and $C\bar{x} = b^T \bar{\pi}$, then $\bar{x}$ is an optimal solution to (P) and $\bar{\pi}$ is an optimal solution to (D).

$$\begin{array}{ll}
 \text{eg } \max & -4x_1 - 2x_2 - x_3 \\
 \text{s.t.} & -x_1 - x_2 + 2x_3 \leq -3 \\
 & -4x_1 - 2x_2 + x_3 \leq -4 \\
 & x_1 + x_2 - 4x_3 \leq 2 \\
 & x_1, x_2, x_3 \geq 0
 \end{array}
 \quad (P) \quad \xleftrightarrow{\text{dual}} \quad
 \begin{array}{ll}
 \min & -3y_1 - 4y_2 + 2y_3 \\
 \text{s.t.} & -y_1 - 4y_2 + y_3 \geq -4 \\
 & -y_1 - 2y_2 + y_3 \geq -2 \\
 & 2y_1 + y_2 - 4y_3 \geq -1 \\
 & y_1, y_2, y_3 \geq 0
 \end{array}
 \quad (D)$$

Given a pair of solutions $\bar{x} = \begin{bmatrix} 0 \\ 4 \\ \frac{1}{2} \end{bmatrix}$ and $\bar{y} = \begin{bmatrix} \frac{9}{2} \\ 0 \\ \frac{5}{2} \end{bmatrix}$ We can justify that $\bar{x}$ is feasible

to (P) and $\bar{y}$ is feasible to (D) Also note that $-4 \times 0 - 2 \times 4 - \frac{1}{2} = -\frac{17}{2} = -3 \times \frac{9}{2} - 4 \times 0 + 2 \times \frac{5}{2}$

We can conclude that $\bar{x}$ is optimal to (P) and $\bar{y}$ is optimal to (D)

• Strong duality

Thm If an LP has an optimal solution, so does its dual, and the respective optimal values are equal

Cor: If both primal and dual are feasible, then they both have optimal solutions and the optimal values are equal.

Summary different possibilities for the primal and the dual

Primal \ Dual	Finite optimum	Unbounded	Infeasible
Finite optimum	✓	✗	✗
Unbounded	✗	✗	✓
Infeasible	✗	✓	✓

• Complementary slackness (CS)

Thm: Let $\bar{x}$ be feasible to (P) and $\bar{\pi}$ be feasible to (D), then $\bar{x}$ is optimal to (P) and $\bar{\pi}$ is optimal to (D) if and only if the following complementary slackness conditions are satisfied.

$$\begin{aligned}\bar{x}_i (A^T \bar{\pi} - c)_i &= 0 & \forall i=1, \dots, n \\ \bar{\pi}_j (b - A \bar{x})_j &= 0 & \forall j=1, \dots, m\end{aligned}$$

eg

$$\begin{array}{ll}\max & -4x_1 - 2x_2 - x_3 \\ \text{s.t.} & -x_1 - x_2 + 2x_3 \leq -3 \\ & -4x_1 - 2x_2 + x_3 \leq -4 \\ & x_1 + x_2 - 4x_3 \leq 2 \\ & x_1, x_2, x_3 \geq 0\end{array} \quad (P) \quad \xleftrightarrow{\text{dual}} \quad \begin{array}{ll}\min & -3y_1 - 4y_2 + 2y_3 \\ \text{s.t.} & -y_1 - 4y_2 + y_3 \geq -4 \\ & -y_1 - 2y_2 + y_3 \geq -2 \\ & 2y_1 + y_2 - 4y_3 \geq -1 \\ & y_1, y_2, y_3 \geq 0\end{array} \quad (D)$$

A pair of optimal solutions are $\bar{x} = \begin{bmatrix} 0 \\ 4 \\ \frac{1}{2} \end{bmatrix}$ and $\bar{y} = \begin{bmatrix} \frac{9}{2} \\ 0 \\ \frac{5}{2} \end{bmatrix}$

$$\bar{x}_1 = 0; (A^T \bar{y} - c)_1 = -\bar{y}_1 - 4\bar{y}_2 + \bar{y}_3 - (-4) = 2 > 0.$$

$$\bar{\pi}_1 (A^T \bar{y} - c)_1 = 0$$

$$\bar{x}_2 > 0; (A^T \bar{y} - c)_2 = -\bar{y}_1 - 2\bar{y}_2 + \bar{y}_3 - (-2) = 0.$$

$$\bar{\pi}_2 (A^T \bar{y} - c)_2 = 0$$

$$\bar{x}_3 > 0; (A^T \bar{y} - c)_3 = 2\bar{y}_1 + \bar{y}_2 - 4\bar{y}_3 - (-1) = 0$$

$$\bar{\pi}_3 (A^T \bar{y} - c)_3 = 0$$

$$\bar{y}_1 > 0; (b - A \bar{x})_1 = -3 - (-\bar{x}_1 - \bar{x}_2 + 2\bar{x}_3) = 0$$

$$\bar{y}_1 (b - A \bar{x})_1 = 0$$

$$\bar{y}_2 = 0; (b - A \bar{x})_2 = -4 - (-4\bar{x}_1 - 2\bar{x}_2 + \bar{x}_3) = 1 > 0$$

$$\bar{y}_2 (b - A \bar{x})_2 = 0$$

$$\bar{y}_3 > 0; (b - A \bar{x})_3 = 2 - (\bar{x}_1 + \bar{x}_2 - 4\bar{x}_3) = 0.$$

$$\bar{y}_3 (b - A \bar{x})_3 = 0$$

Remark: For a feasible solution $\bar{x}$, we call an inequality constraint an active constraint if the constraint is satisfied at equality. The complementary slackness conditions can be rephrased as:

If a primal/dual variable is nonzero, then the corresponding dual/primal constraint is active.

If a dual/primal constraint is inactive, then the corresponding primal/dual variable is zero.